

ARYAN SATYENDRA KUMAR

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EDUCATION

B.Tech in Electronics and Communication Engineering; Minor in Computer Science

National Institute of Technology Goa

CGPA: 8.23/10

2027

Goa, India

Class XII, Goa Board: 85.7%

Loyola Higher Secondary School

2023

Goa, India

EXPERIENCE

Project Intern – Mobile Air Quality Intelligence

Indian Institute of Technology Madras – Prof. Raghunathan Rengaswamy

May 2026 – Jul 2026

Chennai, India

- Developed a multimodal roadside $PM_{2.5}$ modelling pipeline integrating mobile sensors, OPC particle-size bins, MERRA-2/CAMS variables, meteorology, GPS/OSM context, and camera imagery.
- Fine-tuned YOLO vehicle-detection models and used SegFormer road segmentation to extract class-wise traffic counts, road exposure, texture, brownness, dry/gray ratio, and dust-related visual proxies.
- Built ResNet50–GRU temporal and tree-based tabular models for atmospheric-background and local-roadside-increment estimation using leakage-aware whole-date validation.
- Designed a $PM_{2.5}$ filter-sampling and XRF-validation workflow to compare image-derived traffic and road-dust indicators with physical particulate measurements.

Signalling and Telecommunications Intern

South Western Railways, RVNL

May 2025 – Jul 2025

Goa, India

- Studied Kyosan Electronic Interlocking in a safety-critical railway environment, tracing command-to-field feedback paths, interlocking logic, relay interfaces, and failure-mode handling.

PROJECTS

NyayaSetu – ML-RAG Assistant for Legal Aid

FastAPI, Next.js, XLM-R, Sentence Transformers, ChromaDB, Groq API

Jun 2026

GitHub

- Fine-tuned XLM-R classifiers for issue-category and urgency detection across legal-aid, labour, consumer, cybercrime, grievance, and safety queries.
- Built an official-source RAG assistant with ChromaDB retrieval, citation-grounded generation, guardrails, explainability, and evaluation using Recall@1/3/5 and MRR.

F1 Race Predictor – Probabilistic ML Simulation

Python, FastF1, Pandas, NumPy, scikit-learn

Aug 2025 – Present

GitHub | Instagram

- Built an end-to-end ML pipeline using 2023–2025 FastF1 data, leakage-aware rolling-form features, driver/team priors, and circuit context including SC/VSC and pit-loss factors.
- Trained Random Forest models with OOB $R^2 \approx 0.625$, MAE ≈ 2.37 , and RMSE ≈ 3.22 ; added prediction intervals and Monte Carlo race simulations.

HostelMitra – Hostel Complaint Management Portal

Firebase, Firestore, Google OAuth

2025

GitHub | Live Demo

- Built and deployed a full-stack complaint management portal actively used by NIT Goa students, featuring Google OAuth, Firestore persistence, real-time tracking, role-based access, voting, admin resolution, and timestamped audit trails.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++,

ML and Data Science: PyTorch, scikit-learn, Pandas, NumPy, Random Forests, feature engineering, cross-validation, temporal backtesting, uncertainty modelling

Vision, NLP and LLMs: YOLO, SegFormer, CNNs, LSTMs, GRUs, transformers, XLM-R, embeddings, RAG, ChromaDB, retrieval evaluation

Backend and Tools: FastAPI, REST APIs, Firebase, Supabase, PostgreSQL, Firestore, Git, GitHub

Coursework: DSA, Probability and Statistics, Linear Algebra, Numerical Methods, COA, Signals and Systems, Digital Logic

CERTIFICATIONS

Machine Learning Specialization – Andrew Ng | Deep Learning Specialization – Andrew Ng | Google Data Analytics Professional Certificate

ACHIEVEMENTS AND RESPONSIBILITY

- Selected for Amazon ML Summer School 2026, Amazon's competitive program focused on machine-learning foundations and applications.
- Ranked 251 out of 7,000+ teams in the Amazon ML Challenge 2025.
- Finalist in multiple national case-study competitions hosted across IIM, IIT, and NIT platforms.
- Finalist at India Innovates 2026, a national-level hackathon and innovation competition.
- GDG Organizer and AI/ML Lead; led a 100-participant hackathon and conducted ML and DSA workshops.
- Captain, Institute Badminton Team; former national-level badminton player.